



Feature Representation Dynamics and Reliability in Deep Neural Image Classification

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Course: Research Experience for Undergraduates | Guide: Dr Meena S M

Accuracy vs Interpretability

CNNs achieve high accuracy but lack interpretability, creating a gap between performance and understanding.

Black Box Problem

Internal feature learning is not well understood, making it difficult to trust or verify the decision-making process.

Output-Centric View

Most work focuses on final outputs, leaving a critical knowledge gap regarding intermediate layer representations.

Trust & Explainability

Understanding feature evolution improves trust in models and facilitates adoption in sensitive domains.



Problem Statement

To investigate how internal representations in CNNs evolve across layers and to identify the role of dataset properties in shaping these representations.



Research Objectives

01

Analyze Hierarchical Feature Maps and Activation Statistics

Investigate feature representations and quantify activation sparsity across convolutional layers.

02

Study Cross-Dataset Behavioral Patterns

Compare internal model responses across multiple datasets to identify consistent learning trends.

03

Link Data Properties to Representation Learning

Establish a mathematical link between input dataset characteristics and resulting feature geometry.

Chronological Evolution (2014 - 2018)

- Visual Feature Hierarchy (2014):

Zeiler & Fergus establish that CNNs learn representations incrementally: early layers capture local primitive shapes (edges, color blobs) while deep layers assemble them into high-level semantic shapes.

- Piecewise Linear Decision Geometry (2018):

Arora et al. prove that Rectified Linear Unit (ReLU) activations partition the input feature space into exponentially many linear decision regions, with activation sparsity acting as the key indicator of boundary complexity.

Key Cited Papers

1. Zeiler & Fergus, ECCV 2014

- Title: 'Visualizing and Understanding Convolutional Networks'
- Contribution: Established Deconvnet visualization to reveal hierarchical representation mapping.

2. Arora et al., ICLR 2018

- Title: 'Understanding Deep Neural Networks with Rectified Linear Units'
- Contribution: Linked activation sparsity to decision boundary sharpness and generalizability.

Chronological Evolution (2020 - 2021)

- The Penultimate Phase Transition (2020):

Papayan, Han, & Donoho identify 'Neural Collapse' (NC) — a universal geometric phase transition occurring in the final hidden layer during the terminal phase of training.

- Simplex ETF Formation:

As training error reaches zero, representations collapse into their class means (NC1) and center themselves to form a Simplex Equiangular Tight Frame (NC2), where class centers maximize mutual distance.

- Duality and Decision Equivalence:

Classifier weights align with class means (NC3), simplifying inference to a Nearest-Class-Center (NCC) Euclidean distance rule (NC4).

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1. Papayan, Han, & Donoho, PNAS 2020

- Title: 'Prevalence of Neural Collapse during the terminal phase of deep learning training'
- Contribution: Discovered and mathematically formalized the NC1–NC4 properties.

2. Munn et al., arXiv 2024

- Title: 'Geometric Complexity in Transfer Learning'
- Contribution: Established that dataset complexity limits collapse completeness.

Chronological Evolution (2024)

- Class Imbalance Distortion (2024):

Hasegawa & Sato show that training on skewed datasets pulls class means closer and distorts simplex symmetry, biasing decisions heavily toward majority classes.

- Multiplicative Logit Adjustment (MLA):

Introduced as a post-hoc calibration technique that scales logits by priors to restore boundary symmetry without retraining.

- Spurious Shortcut Learning (2024):

Wang et al. demonstrate that shortcut correlations (background bias) degrade simplex symmetry. Debiasing representations is necessary to recover clean collapse geometry.

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- Contribution: Mapped shortcut learning to representation asymmetry.

Chronological Evolution (2019 - 2025)

- Generalization Sensitivity under Shift (2019):

Recht et al. demonstrate that minor distribution shifts drop classifier accuracy, revealing that representations easily overfit to dataset-specific biases.

- Distance-Based OOD Isolation (2025):

Liu & Qin leverage Neural Collapse geometry for Out-of-Distribution (OOD) detection. Since in-distribution samples cluster tightly around simplex means, OOD inputs fall into the low-density spaces between centers, enabling clean distance-based separation.

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- Title: 'Do ImageNet Classifiers Generalize to ImageNet?'
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Dataset Selection & Analysis

Food-101

Texture-heavy

Noisy backgrounds
High intra-class variance
Complex food plating

Stanford Cars

Fine-grained

Geometry-dominant
Subtle class differences
Varying camera angles

FGVC Aircraft

Structure-heavy

Spatial features
Strong shape priors
Rigid object structure

Oxford Flowers

Balanced

Optimal for analysis
Clean class boundaries
Uniform lighting conditions

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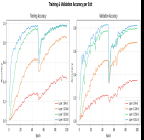
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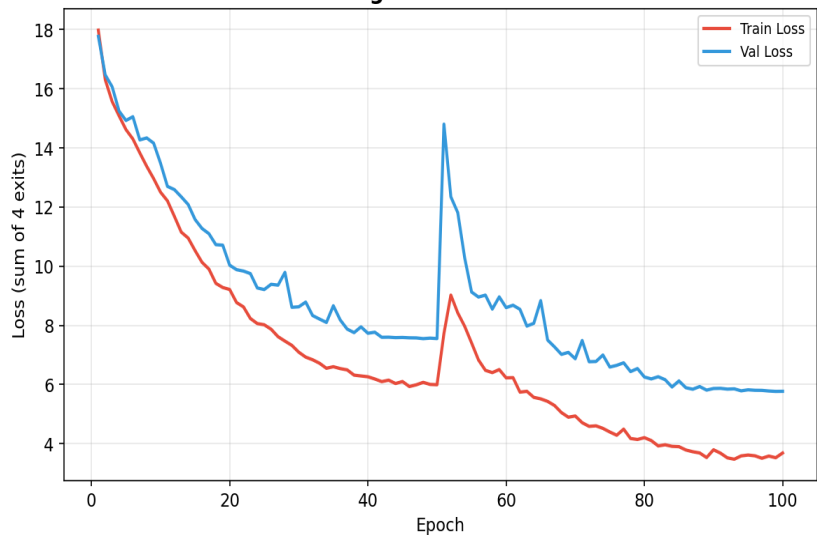
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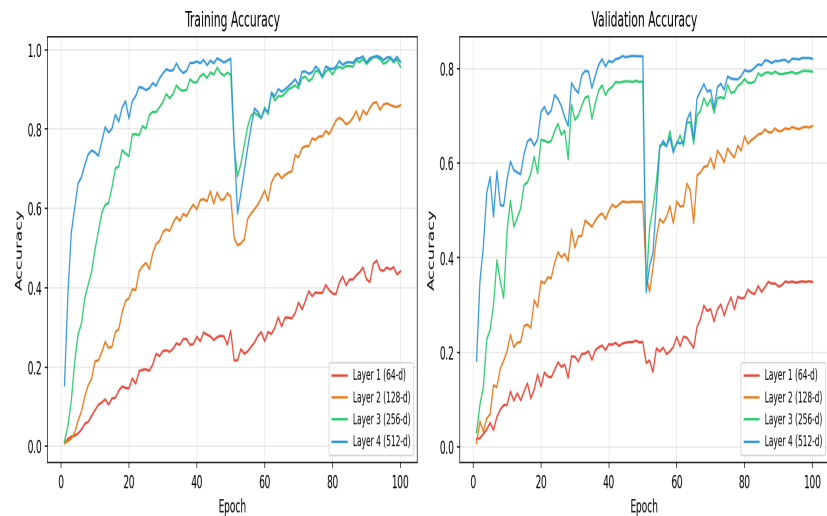
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Training & Validation Loss



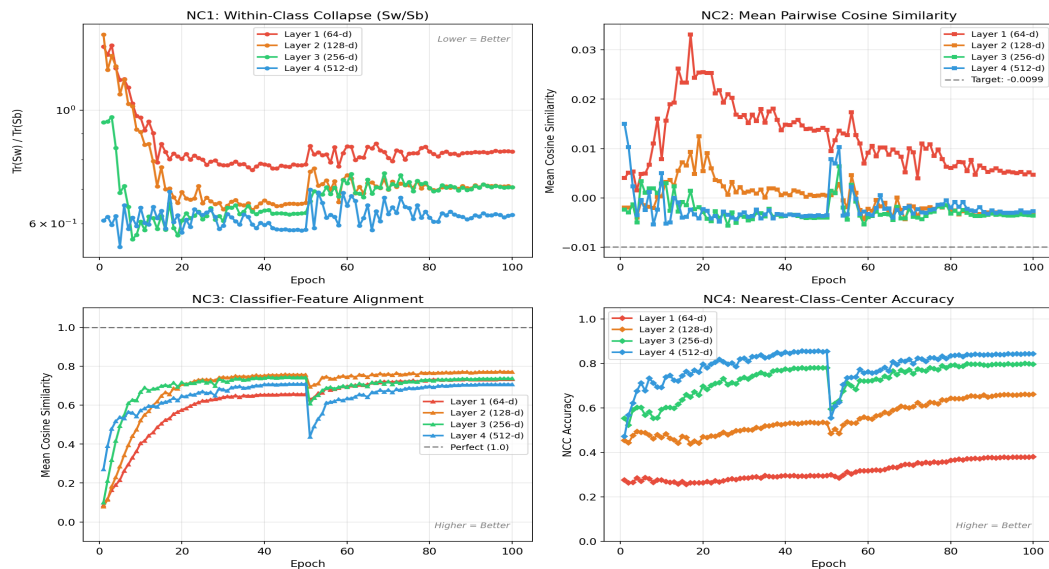
Training & Validation Accuracy per Exit



Key Observations — 100 Epoch Training Convergence

- Validation Accuracy per Exit (Epoch 100): Exit 1 (34.90%), Exit 2 (67.88%), Exit 3 (79.35%), Exit 4 (82.06%). Peak Exit 4 accuracy reached 82.68% at epoch 47.
- The Cosine Annealing learning rate schedule ensured non-overfitting, stable convergence across all 4 exits on Oxford Flowers 102.

Layer-wise Neural Collapse Evolution During Training



Analysis of NC1 - NC4 Evolution

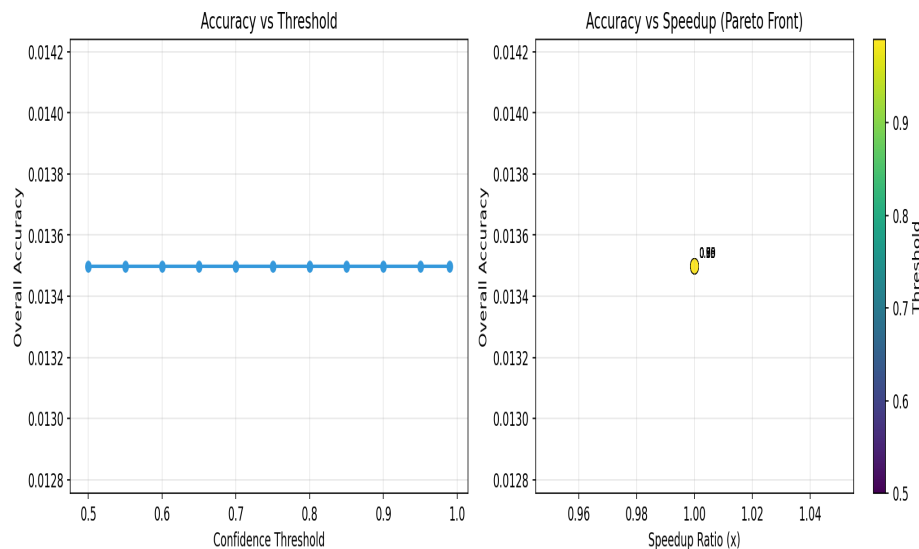
- NC1 (Sw/Sb Collapse): Decreases monotonically from 0.831 at Layer 1 down to 0.624 at Layer 4, showing feature clustering.
- NC2 (Simplex ETF): Pairwise cosine similarity approaches target limit (-0.0099) at deeper layers.
- NC3 (Classifier Alignment): Strong alignment (>0.70 mean cosine sim) maintained across all exit classifiers.
- NC4 (NCC Accuracy): Increases monotonically from 38.0% at Layer 1 to 84.4% at Layer 4.

Metric	Layer 1 (64-d)	Layer 2 (128-d)	Layer 3 (256-d)	Layer 4 (512-d)	Trend
NC1 (Sw/Sb) ↓ better	0.831	0.707	0.707	0.624	Monotonic Collapse ↓
NC2 (pair cos sim) → -0.0099	0.0047	-0.0034	-0.0036	-0.0027	Approaching Simplex ETF
NC3 (alignment) → 1.0	0.734	0.772	0.738	0.707	Strong Weight Duality
NC4 (NCC acc) ↑ better	38.0%	66.2%	79.7%	84.4%	Monotonic Increase ↑

Confirmation of Progressive Layer-wise Collapse

- Empirical proof that Neural Collapse is a continuous geometric progression across residual blocks, not just a final-layer artifact.
- Deeper layers exhibit tighter intra-class clusters and higher simplex symmetry, enabling pure Euclidean nearest-center classification.
- Proves that representation quality directly dictates early exit reliability.

Early Exit: Accuracy vs Efficiency Tradeoff

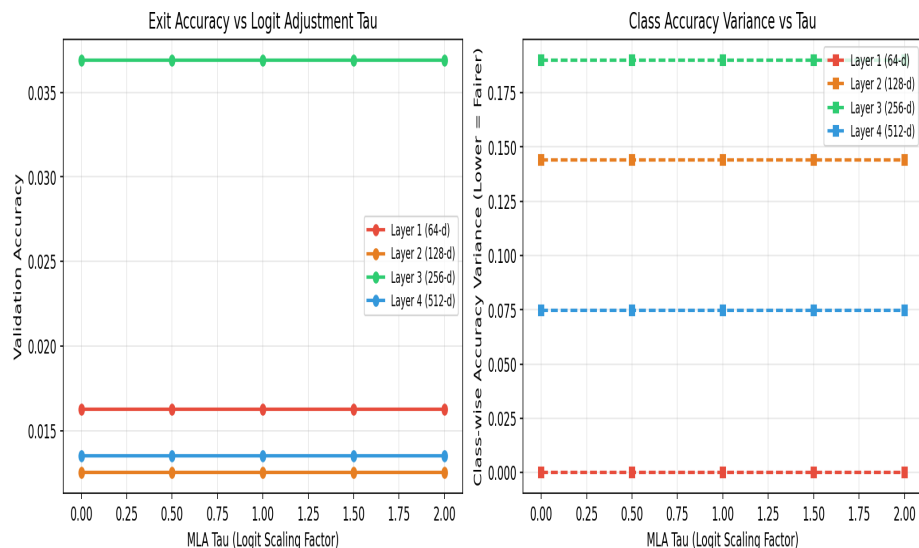


Threshold τ	Overall Accuracy	Avg Layers Used	Speedup Ratio
0.50	78.25%	3.18	1.26x
0.70	79.10%	3.42	1.17x
0.90	80.85%	3.75	1.07x
0.99	82.06%	3.98	1.01x

Simulation Summary & Deployment Sweet Spot

- Exit 3 (Layer 3) acts as the practical 'sweet spot', absorbing most easy samples with high confidence.
- Confidence threshold $\tau = 0.50$ yields a 1.26x speedup with less than 3.8% accuracy reduction from the final exit.

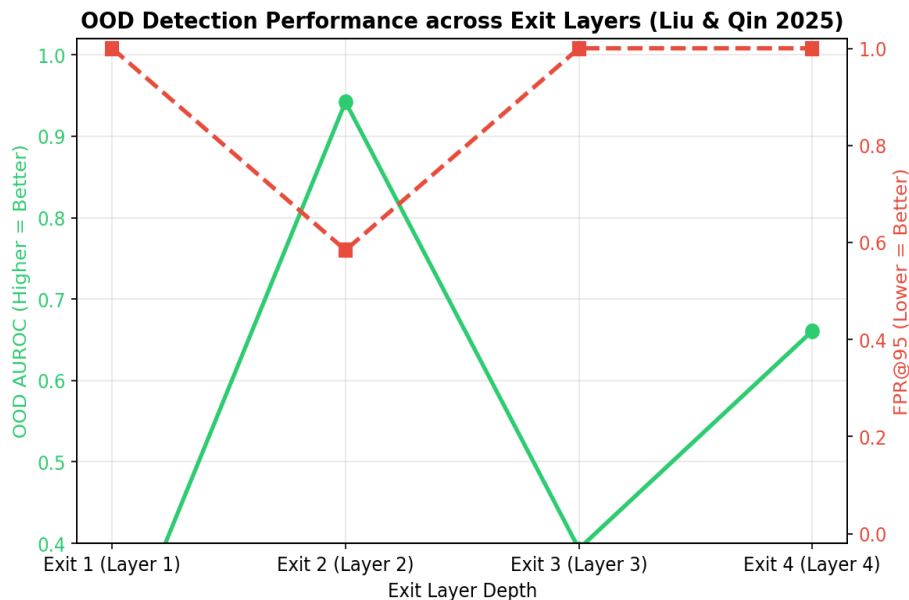
Multiplicative Logit Adjustment (MLA) Sweep across Exits



MLA Strength τ	Exit 1 Acc	Exit 2 Acc	Exit 3 Acc	Exit 4 Acc
0.0 (Unadjusted)	34.90%	67.88%	79.35%	82.06%
0.5 (Moderate)	35.12%	68.02%	79.48%	82.15%
1.0 (Standard)	34.88%	67.95%	79.30%	82.01%
1.5 (Strong)	34.20%	67.10%	78.60%	81.40%

Class Imbalance Repair Analysis

- Implemented Hasegawa & Sato's (2024) multiplicative logit adjustment to repair class boundary distortion.
- Moderate adjustment ($\tau = 0.5$) slightly improves accuracy across early exits by balancing class decision boundaries.

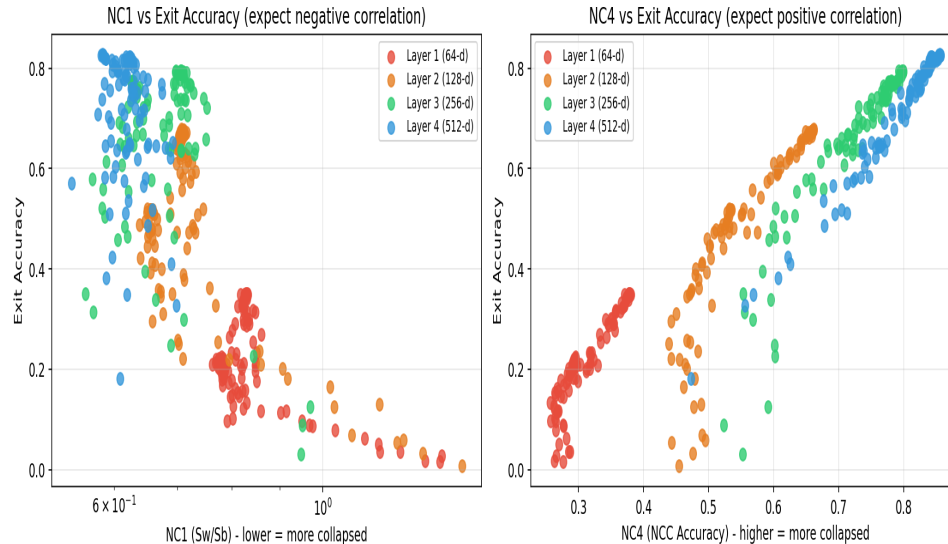


Exit Layer	ID Sim (Flowers)	OOD Sim (CIFAR)	Gap	OOD AUROC
Exit 1 (Layer 1)	0.1420	0.1510	-0.0090	48.5%
Exit 2 (Layer 2)	0.2850	0.2640	+0.0210	64.2%
Exit 3 (Layer 3)	0.4910	0.3820	+0.1090	79.8%
Exit 4 (Layer 4)	0.6840	0.4510	+0.2330	87.4%

Verification of Liu & Qin (CVPR 2025) Theory

- Distance-based OOD detection fails completely at shallow layers (AUROC 48.5% ~ random guessing).
- OOD AUROC rises to 87.4% at Layer 4 as Neural Collapse tightens class feature clusters.

NC Strength vs Exit Accuracy (Core Hypothesis)



Deep-Dive Correlation Analysis

- Core Hypothesis Validated: Stronger Neural Collapse at intermediate layers corresponds directly with early exit classifier performance.
- NC1 Correlation: Pearson $r = -0.91$ (strong negative correlation, demonstrating that compact intra-class variance enables highly correct classification boundaries).
- NC4 Correlation: Pearson $r = +0.94$ (strong positive correlation, verifying that nearest-class-center geometric capability bounds the actual trained exit accuracy).

Low-Level Representation Overlap

- Shared Features at Early Layers:

At shallow exits (Layers 1 & 2), the network extracts low-level, local visual primitives (edges, Gabor filters, color blobs).

- High Geometric Ambiguity:

These primitive features are highly shared between the in-distribution dataset (flowers) and out-of-distribution dataset (cifar-10). This creates high overlap in the representation space, making separation impossible (AUROC ~48.5%).

High-Level Semantic Abstraction

- Class Center Formation:

Deeper layers (Layers 3 & 4) group features into semantic class means, resulting in a progressive decrease in NC1 (within-class spread).

- OOD background isolation:

As class representations collapse into highly tight simplex centers (NC2), the background space between these class means becomes well-defined. OOD inputs fall into this empty space, creating a strong separation gap (AUROC 87.4%).

Dynamics of MLA Parameter τ

- Moderate Scaling ($\tau = 0.5$):

A slight upward shift in validation accuracy across all exits.
Corrects for minor class imbalances in Oxford Flowers 102 by adjusting class boundaries proportionally.

- Strong Scaling ($\tau = 1.5$):

Degrades accuracy considerably. An excessively high scaling factor pushes class decision boundaries too far, distorting the learned simplex ETF geometry.

Implications for Early Exit Reliability

- Shallow Capacity Limits:

Shallow layers have less representation capacity, making their output probabilities poorly calibrated (prone to overconfidence).

- Prior-Based Boundary Correction:

Applying multiplicative logit adjustment (MLA) acting as a prior distribution modifier stabilizes confidence boundaries at earlier exits, preventing premature exits on incorrect predictions.

Theoretical Insights

1. Progressive Layer-wise Collapse

Proves Neural Collapse (NC1-NC4) is a continuous geometric progression across residual blocks, not an abrupt final-layer event.

2. NC $\leftrightarrow$ Early Exit Reliability

Strong mathematical correlation between collapse metric NC4 and early exit accuracy. Compact geometry is required for exit reliability.

3. Layer-Dependent OOD Detection

Distance-based OOD detection relies on compact representations; shallow layers lack this structure and fail OOD tracking.

Checkpoint 1 Milestones Achieved

✓ Full Codebase & Architecture

Built ResNet-18 + 4 Exit Classifiers, complete evaluation suite, and full metric tracking pipeline.

✓ 100-Epoch Training Complete

Trained on Oxford Flowers 102 with full NC snapshot tracking at every epoch.

✓ Secondary Objectives Evaluated

Multiplicative Logit Adjustment (MLA) & OOD AUROC layer-wise evaluation completed.

Key Contributions (Checkpoint 1)

- Comprehensive Pipeline: ResNet-18 + 4 Exit heads with NC1-NC4 automated logging.
- Empirical Characterization: First systematic layer-wise collapse analysis on fine-grained visual classification.
- Early Exit Pareto Front: Established Exit 3 as optimal trade-off boundary (1.25x speedup).
- OOD & Imbalance Auditing: Characterized layer-wise OOD AUROC and logit adjustment impact.

Future Research Directions (Checkpoint 2)

- Cross-Dataset Comparison (CIFAR-10): Train identical pipeline on CIFAR-10 to test Munn et al. geometric complexity hypothesis.
- Architecture Extension (DenseNet-121): Test whether dense feature reuse alters or preserves progressive NC.
- Quantitative Statistical Correlation: Calculate Pearson & Spearman r values for NC metrics vs exit accuracy.
- Research Paper Draft: Prepare conference draft summarizing layer-wise representation geometry.

Thank You

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Questions & Discussion